

RAG BASED MACHINE LEARNING ASSISTANT

The aim of this assignment is to create a RAG based system to get relevant information from a Machine learning textbook

Embedding Model

The project used HuggingFaceEmbeddings with the Sentence Transformer model:all-MiniLM-L6-v2

This model converts text into dense vector representations (embeddings) that capture the semantic meaning of the text. Semantically similar texts produce similar vectors, enabling efficient similarity-based retrieval.

The embedding model was selected because it is:

- Lightweight and fast
- Open-source and free to use
- Suitable for semantic search tasks
- Efficient for CPU-based environments
- Widely used in Retrieval-Augmented Generation (RAG) systems

The model was integrated using LangChain and HuggingFace embeddings.

Vector Database

The project used FAISS as the vector database for storing and retrieving embeddings.

FAISS stores vector representations of document chunks and performs efficient similarity search using vector distance calculations.

FAISS was selected because it provides:

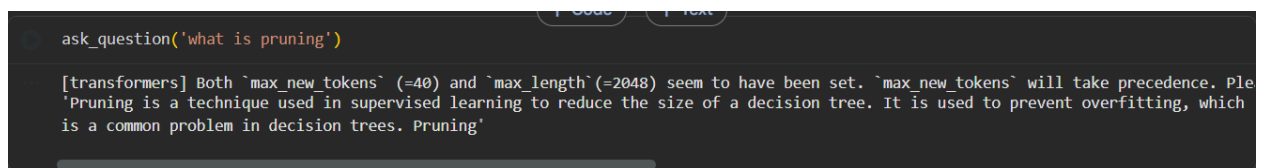
- Fast similarity search
- Efficient vector indexing
- Low memory usage
- Scalability for large datasets
- Easy integration with LangChain

Prompt Design

The prompt was designed with the following goals:

- Reduce hallucinations
- Restrict the model to retrieved content
- Generate concise and relevant answers
- Improve grounding and factual consistency

The prompt explicitly instructs the model to answer only using the provided context and avoid generating unsupported information.



```
ask_question('what is pruning')

[transformers] Both `max_new_tokens` (=40) and `max_length` (=2048) seem to have been set. `max_new_tokens` will take precedence. Please refer to this page on Hugging Face: https://huggingface.co/docs/transformers/main/en/generation_strategies#truncate-after-n-tokens
'Pruning is a technique used in supervised learning to reduce the size of a decision tree. It is used to prevent overfitting, which is a common problem in decision trees. Pruning'
```

Answer generation

The retrieved context is passed to a language model: TinyLlama/TinyLlama-1.1B-Chat-v1.0.

The RAG system receives query retrieves relevant contents and generates answer.

Multiple queries were tested.

Result

The project successfully implemented a functional Retrieval-Augmented Generation (RAG) application capable of answering Machine Learning and Deep Learning related questions using a PDF-based knowledge source.

The application integrates the following components into a complete end-to-end pipeline:

- PDF document loading
- Text extraction
- Text chunking
- Embedding generation
- Vector database storage
- Semantic retrieval
- Context-aware answer generation

The system accepts user queries in real time, retrieves the most relevant document chunks using semantic similarity search, and generates grounded answers using a language model.

 ask_question('what is overfitting')

[transformers] Passing `generation\_config` together with generation-related arguments=({'do_sample', 'max_new_tokens'}) is deprecated.
[transformers] Both `max\_new\_tokens` (=40) and `max\_length` (=2048) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the model documentation to avoid overfitting. Overfitting occurs when a model fits the training data too closely, leading to a model that works well on the training data but is not able to generalize to new data.
Long

 ask_question('what is pruning')

[transformers] Both `max\_new\_tokens` (=40) and `max\_length` (=2048) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the model documentation to avoid overfitting. Pruning is a technique used in supervised learning to reduce the size of a decision tree. It is used to prevent overfitting, which is a common problem in decision trees. Pruning